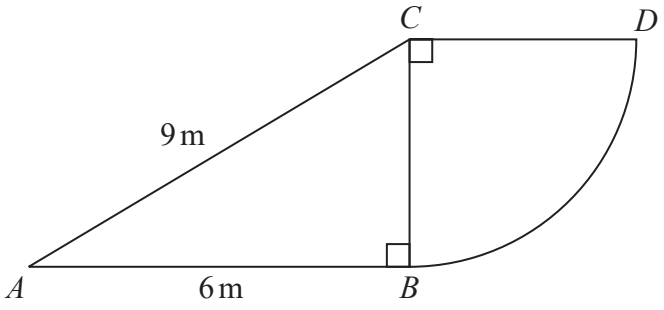
27 The diagram shows a right-angled triangle and a quarter circle.



The right-angled triangle ABC has angle $ABC = 90°$
The quarter circle has centre C and radius CB.

Work out the area of the quarter circle.
Give your answer correct to 3 significant figures.
You must show all your working.

Right triangle: $CB^2 = AC^2 - AB^2 = 9^2 - 6^2$.

$CB^2 = 81 - 36 = 45$ (so $CB = \sqrt{45}$ m).

Quarter-circle area = $\pi r^2/4 = 45\pi/4$.

$45\pi/4 = 35.3429... \approx 35.3$ m^2 (3 s.f.).

4 marks total
Pythagoras set-up
Radius^2 = 45
$\pi r^2/4$
35.3 m^2

..... m^2

(Total for Question 27 is 4 marks)

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